

CAD Laboratory (CE4P001) — Assignment 2

Session: Autumn 2025

Report

by

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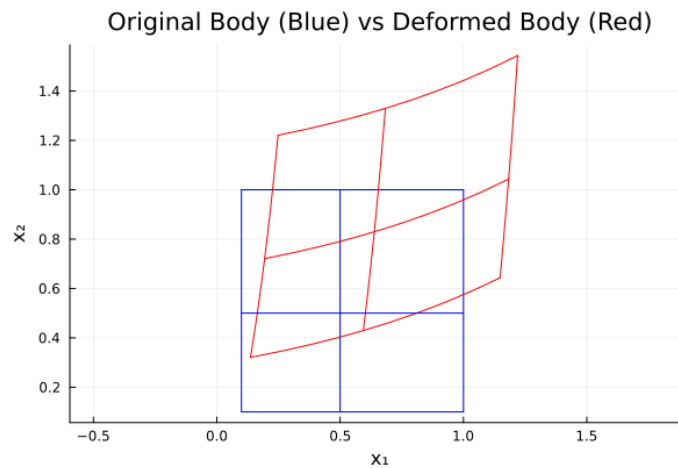
Indian Institute of Technology, Bhubaneswar

Odisha

November 2025

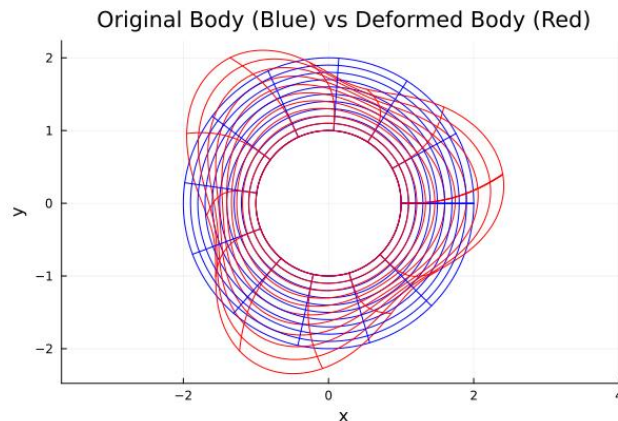
Question 1: Displaced Shape of 2D Body

- Using Plots.jl is imported to generate 2D plots.
- x_1 and x_2 grids are generated using step ranges and broadcasted to form a mesh.
- Displacement components u_1 and u_2 are computed using the logarithmic and exponential functions.
- The deformed coordinates $x_1 + u_1$ and $x_2 + u_2$ are obtained.
- Boundary lines and internal grid lines are plotted for both original (blue) and deformed (red) configurations.



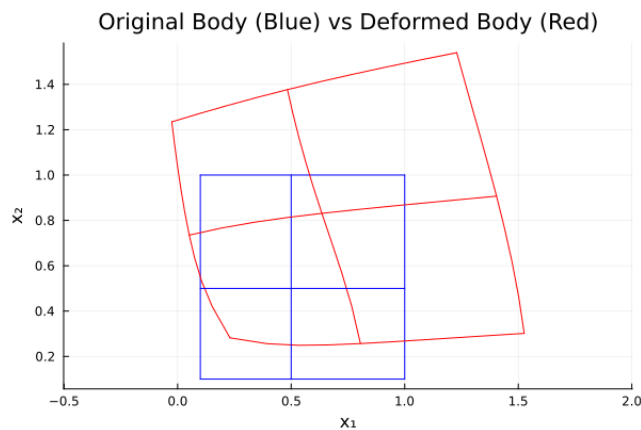
Question 2: Annular Body in Polar Coordinates

- Radial (r) and angular (θ) grids are defined using uniform sampling.
- Converted (r, θ) to Cartesian coordinates (x, y).
- Defined deformation fields u_r and u_θ based on the given expressions.
- Converted deformation from polar to Cartesian components (u_x, u_y).
- Computed deformed coordinates $x+u_x$ and $y+u_y$ using broadcasting.
- Plotted radial and circular lines before and after deformation.



Question 3: Deformation with Polar Transformation

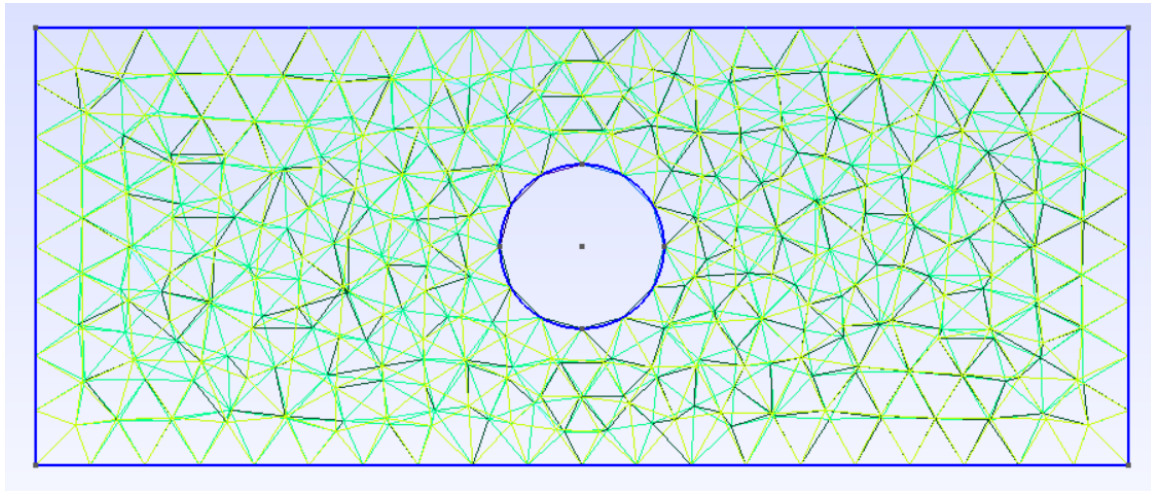
- Generated rectangular grid (x_1, x_2).
- Converted grid points to polar radius r and angle θ .
- Defined u_r and u_θ based on exponential and logarithmic terms.
- Converted polar displacement to Cartesian displacement (u_1, u_2).
- Added displacement to original coordinates and plotted.



Question 4: Plate with Circular Hole — Meshing

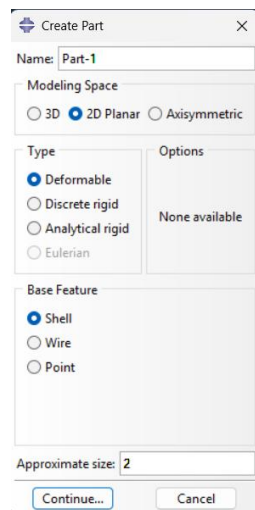
- Defined geometric parameters L , H , thickness and hole radius.
- Added corner points and hole center/arc points in Gmsh.
- Created outer rectangular loop and circular inner hole loop.
- Plane surface defined with hole subtraction.
- Extruded surface to 3D to obtain plate volume.
- Assigned physical groups for boundaries and volume.
- Generated 3D mesh.
- Governing PDE: $\nabla \cdot \sigma = 0$, $\varepsilon = (\nabla u + \nabla u^T)/2$, $\sigma = 2\mu \varepsilon + \lambda \text{tr}(\varepsilon)I$.
- Boundary conditions: left edge fixed, all other edges traction-free except loaded side

Julia Meshing:

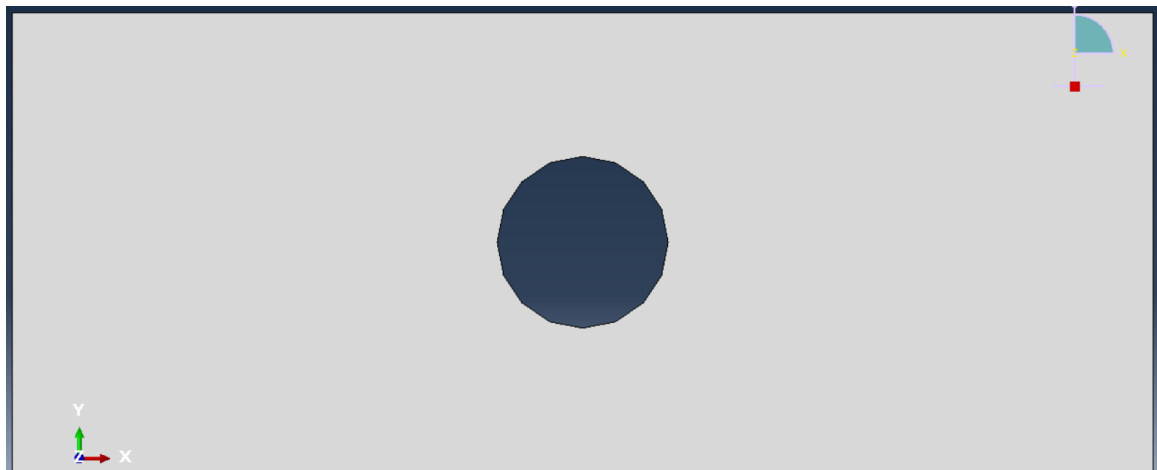


Abaqus Meshing:

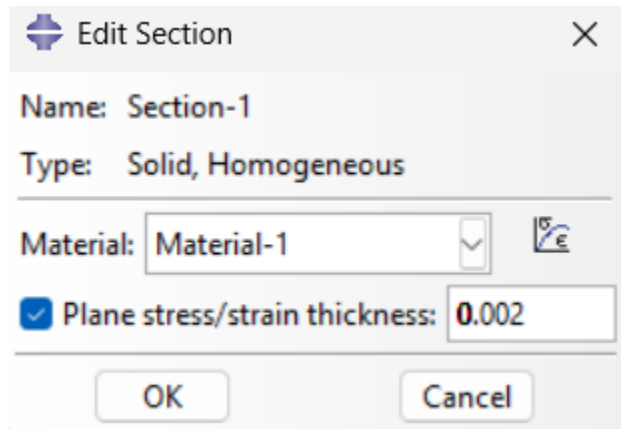
1) A 2D shell part was created.



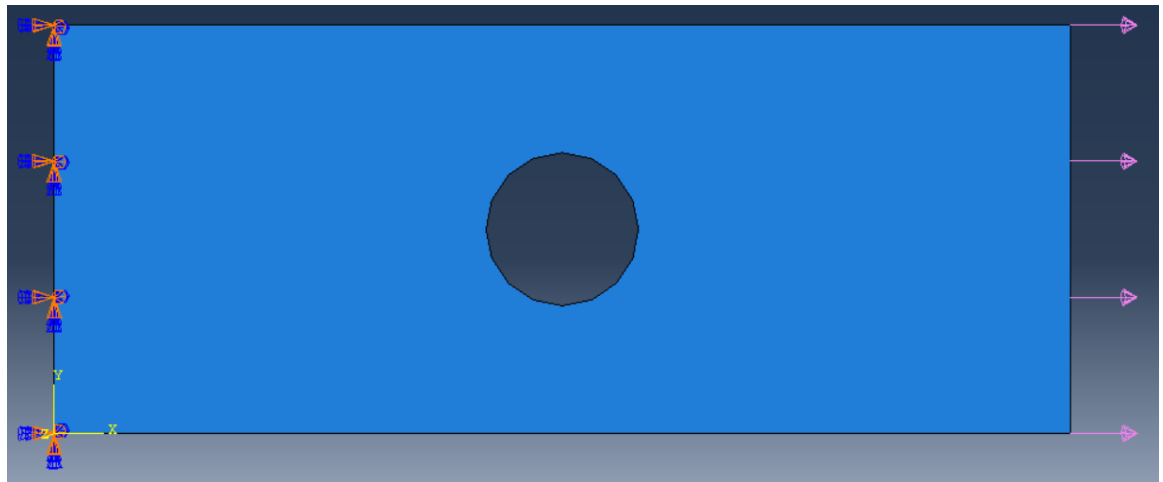
2) Geometry was defined.



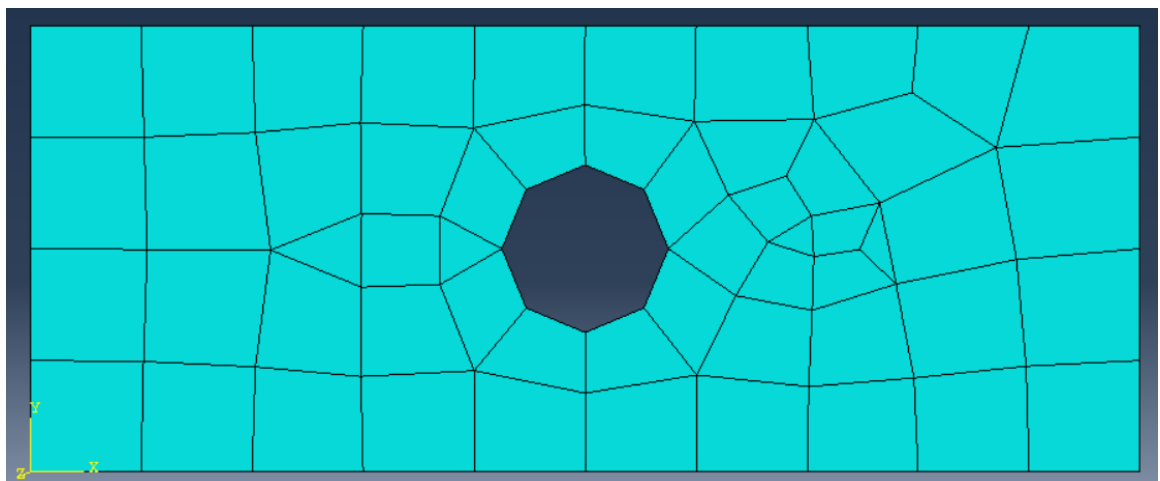
- 3) With the given elasticity and Poisson's ratio, material was created, followed by assigning of section with planar stress thickness 2×10^{-3} m (everything was considered in m compatible units). This was followed by assembly and assigning steps.



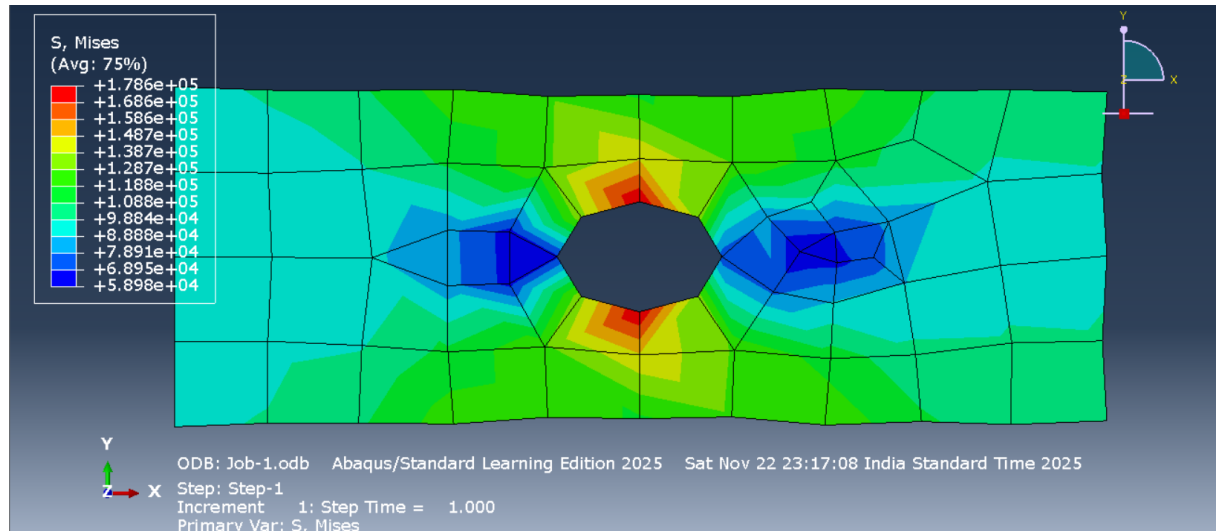
- 4) A pressure load of 100×10^3 N/m was applied and fully constrained encastered boundary condition was set.



- 5) Generated mesh.



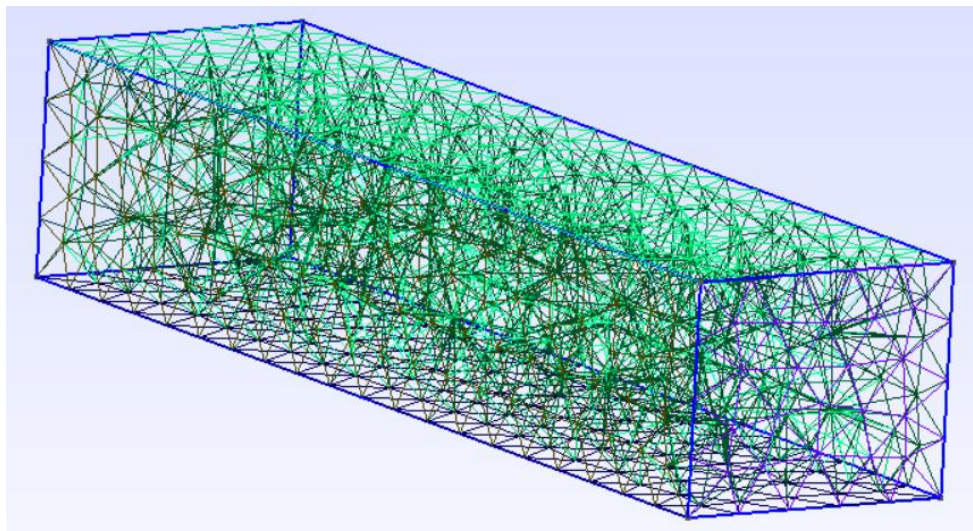
6) Job was created and results for Von-Mises stress was visualized.



Question 5: Cantilever Beam with Point Load — Meshing

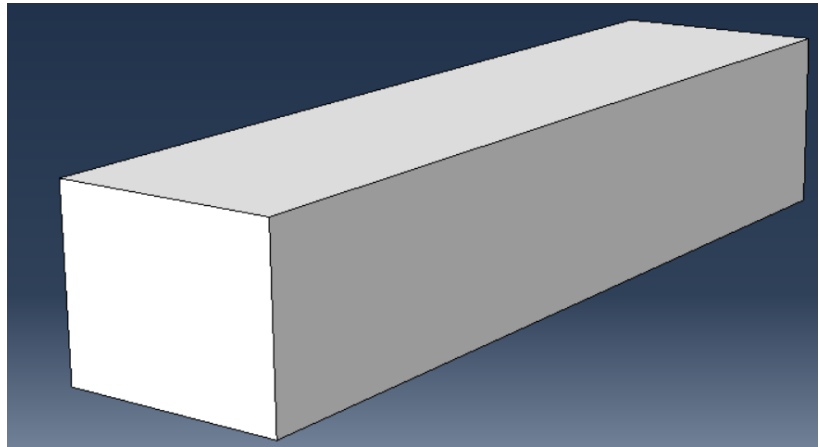
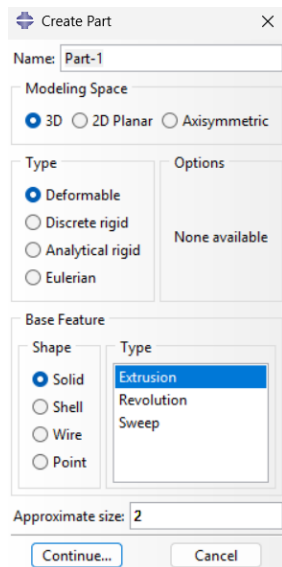
- Defined beam dimensions l , w , d .
- Added rectangular cross-sectional points in Gmsh.
- Created lines and surface loop.
- Extruded surface to form cantilever volume.
- Generated mesh.
- Governing PDE same as Q4 (linear elasticity).
- Boundary: fixed at one end; point load applied at free end.

Julia Meshing:

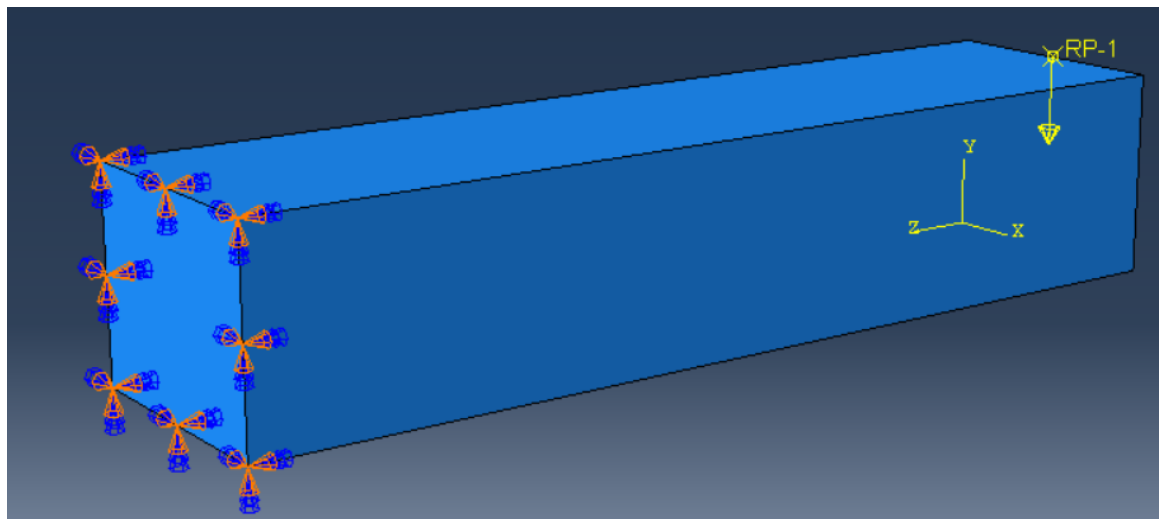


Abaqus Meshing:

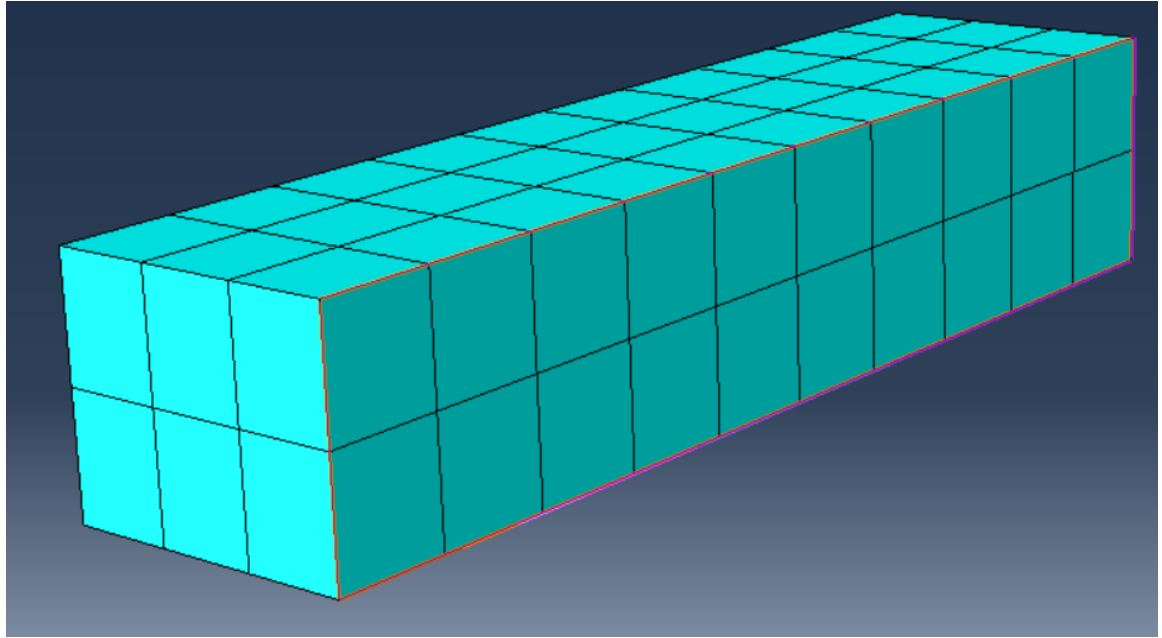
- 1) All steps are same as above, but the part was defined to be as a 3D.
Geometry was defined, material properties were given ($E = 25\text{GPa}$, $\nu = 0.2$) and section was defined followed by creation of assembly and steps



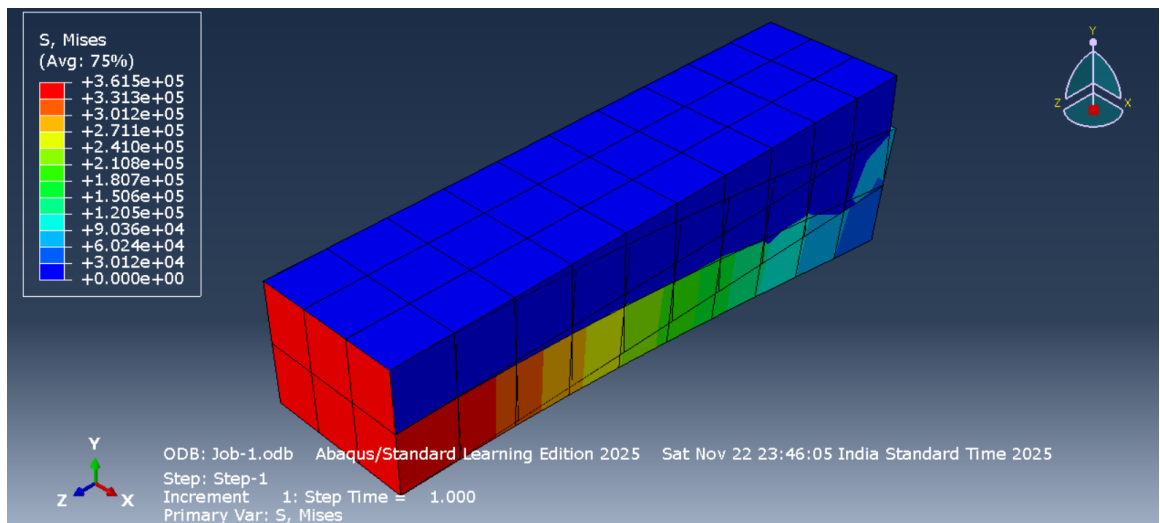
- 2) In Load module, a reference point was created (also this reference was tied with the same edge in interaction module) for the application of concentrated load at center of right edge and left surface was completely fixed by encastre.



- 3) Global seeds were assigned and mesh was generated.



4) Job was created and results for Von-Mises stress were visualized.

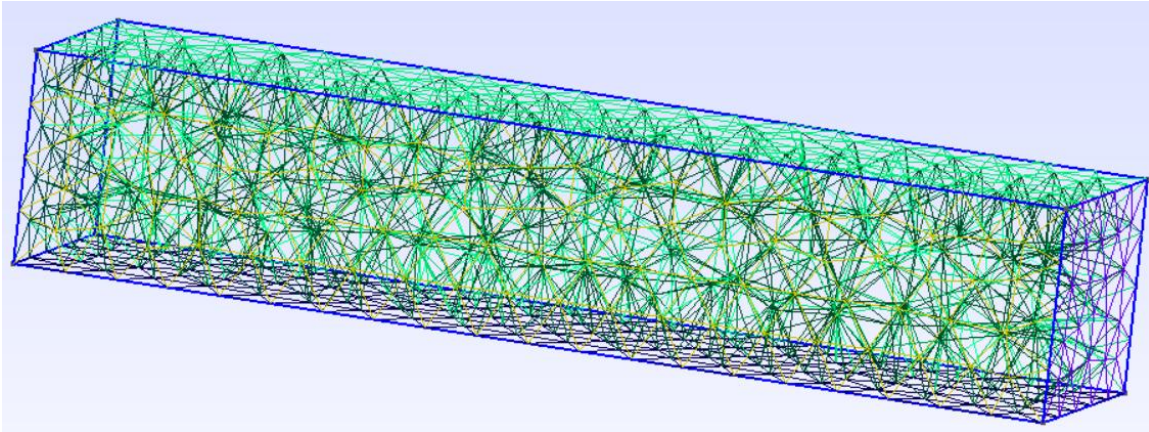


Question 6: Cantilever Beam with UDL — Meshing

- Same geometry construction as Question 5 but for UDL case.
- Generated 3D mesh using Gmsh extrusion.
- Assigned boundary groups.

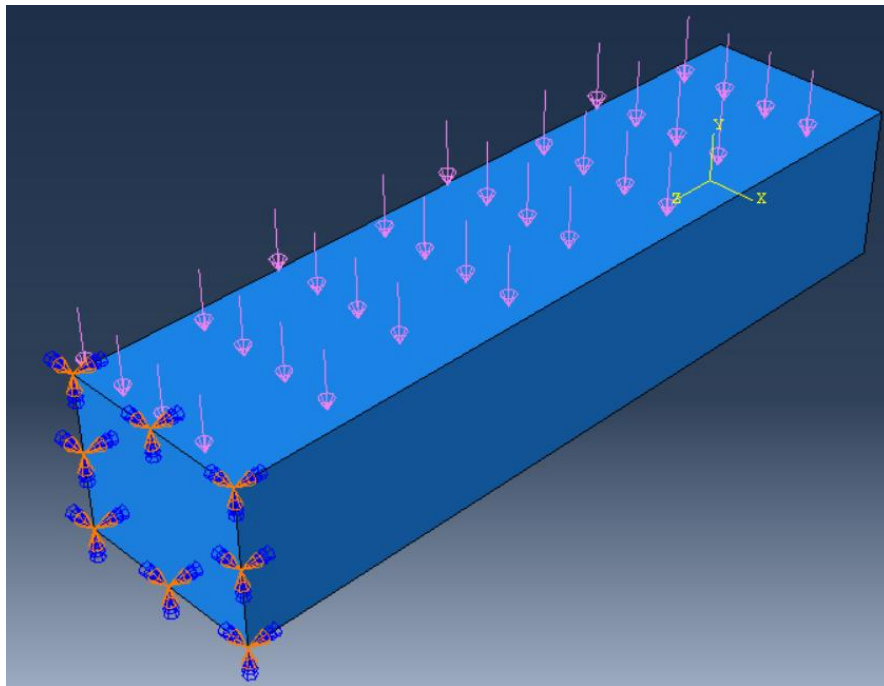
- PDE: linear elastic solid under distributed load.
- Boundary: fixed end + UDL on top face.

Julia Meshing:

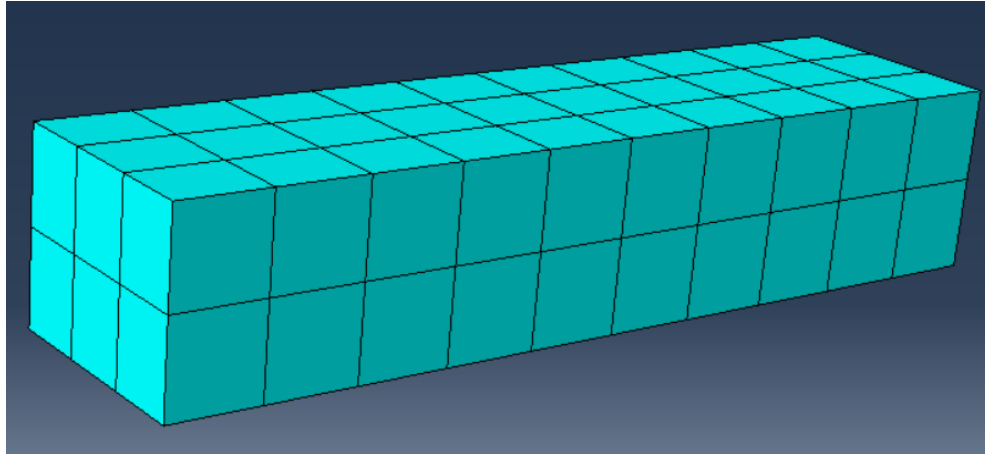


Abaqus Meshing:

- 1) Same steps as in Question 5, only load here is, 1GPa pressure load on top surface.



- 2) Global seeds were set and mesh was generated



3) Job was created and results for Von-Mises stress were visualized.

